

LLM Driven Full End to End PCB Generation for KiCAD

TechBlazers 2.0 : Project Pre Demo

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What We Built

Goal: a plain-language circuit description → a manufacturable KiCad schematic, fully automatically, for STMicroelectronics parts.

The flow, end to end

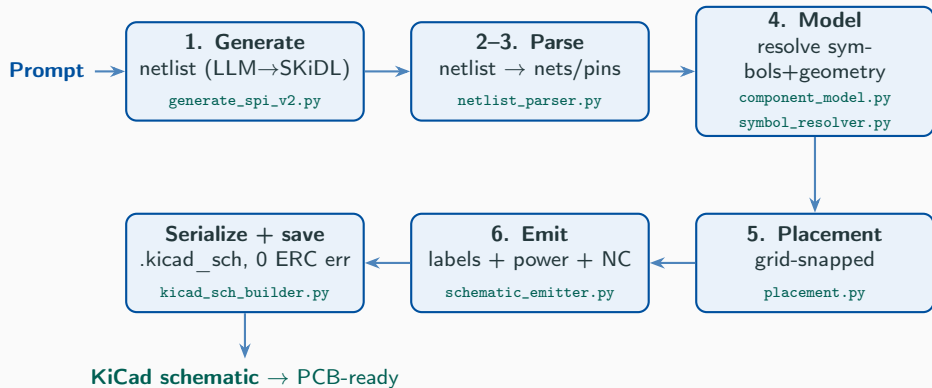
- ▶ Natural-language prompt
- ▶ → ERC-validated SKiDL netlist
- ▶ → native KiCad 9 .kicad_sch
- ▶ → footprints assigned → PCB-ready

Validated benchmark

- ▶ STM32L476JGY + LSM6DSM over SPI
- ▶ 25 components placed automatically
- ▶ **0 ERC errors** in KiCad
- ▶ Both ICs place on PCB with correct footprints

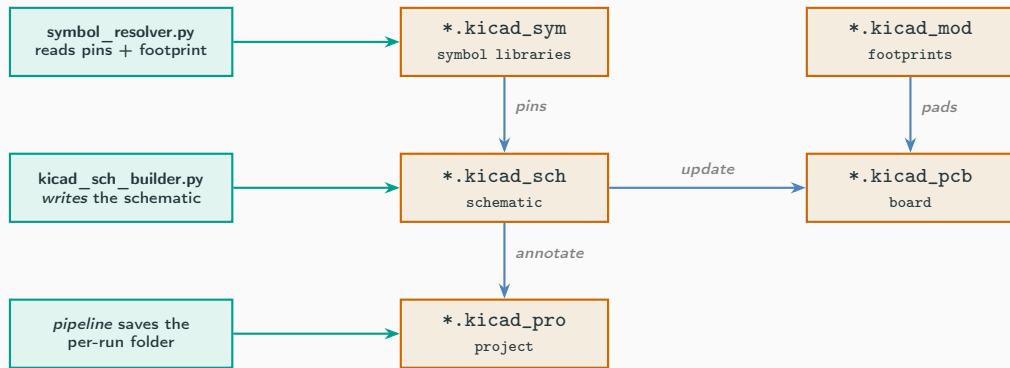
*Two correctness problems kept separate throughout: **netlist correctness** (the LLM's job) vs. **schematic-file correctness** (the serializer's job).*

Our Pipeline — One Command, Six Stages



Orchestrated end-to-end by `pipeline.py`; driven from a Streamlit UI (`ui.py`). Datasheet RAG context (`build_rag_v2.py`, `pymupdf4llm`) feeds Stage 1.

How KiCad Builds a Schematic — and Where We Plug In



- Our pipeline ■ KiCad-owned files. The symbol library is our *authoritative source* for pins + footprints (no LLM hallucination); we *generate* the `.kicad_sch` directly and leave the project & board files to KiCad.

Inside the Intermediate File: .kicad_sch

KiCad's schematic is a human-readable **S-expression** text file. Our serializer emits it directly. Key structures we write:

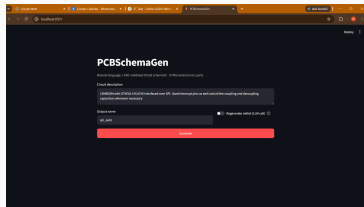
```
(kicad_sch (version ...) (uuid <root>)  
  (lib_symbols  
    (symbol "Device:C" ...)) % embedded  
    (label "SPI_SCK" (at X Y)) % net connect  
    (symbol (lib_id "power:GND") ...)  
    (symbol (lib_id "MCU_ST_STM32L4:...")  
      (property "Footprint" "...")  
      (instances (project ...  
        (path "<root>" ...))))))
```

What lives inside & how we use it

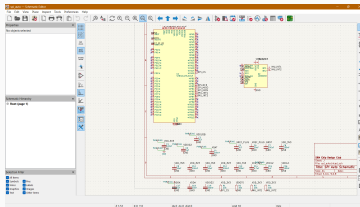
- ▶ **lib_symbols** — authoritative symbol text sliced from KiCad's library (byte-matches $\Rightarrow$ no mismatch warnings).
- ▶ **labels** — same-named labels connect nets; how we wire without routing.
- ▶ **instance path UUID** — must equal the root sheet UUID or KiCad shows all refs as "?".
- ▶ **Footprint property** — read authoritatively from the symbol $\Rightarrow$ ICs place on the PCB.

Enhancement path: parse these same structures back to auto-score generated schematics against ST reference boards.

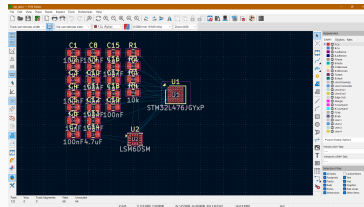
Outputs



Interactive UI (ui.py)



Generated schematic, 0 ERC errors



PCB update: ICs placed

Tools & Software

Category	Tool	Role in the project
LLM	Claude (Sonnet)	Netlist design from prompt
Netlist	SKiDL	Programmatic netlist + ERC
EDA	KiCad 9	Symbols, schematic, PCB, ERC
KiCad CLI	kicad-cli	Headless SVG schematic render
Symbols	kiutils	Parse .kicad_sym geometry
RAG store	ChromaDB	Datasheet vector retrieval
Extraction	pymupdf4llm	Layout-aware datasheet → markdown
Embeddings	sentence-transformers	all-MiniLM-L6-v2
Chunking	langchain-text-splitters	Markdown-aware chunking
UI	Streamlit	Interactive generation front-end
Runtime	Python 3.10 / WSL Ubuntu	Environment
Docs	L ^A T _E X / Beamer	This deck